

Deconstructing HCI & Computer Graphics in Daily Software

Abdul Khaliq · Roll Number: 2K24-CSE-6

Submitted to: Sir Rajesh Kumar · Introductory HCI & Computer Graphics · September 10, 2026

App analyzed: Google Earth (mobile app) – Eiffel Tower, Paris. **Screenshots:** hci_screenshot . jpg (tilted 3D view: search bar, compass, ruler tool) and cg_screenshot . jpg (top-down view: mesh geometry, LOD % indicator).

Part 1 – HCI Deconstruction



1. Search Bar

Sits at the top and is the main entry point for navigation – type a place name and the camera flies there automatically. It hides camera-positioning complexity (lat/long, altitude, tilt) behind a plain text box, so the user never has to think in 3D coordinates, which keeps cognitive load low.

2. Compass / Tilt Gizmo

Shows current orientation and lets the user tap/drag to reset the view to north. It's a direct-manipulation control – feedback is immediate, matching the user's mental model of “grabbing” the world. Without it, re-orienting after a free pan would require guesswork.

3. Ruler Tool

The ruler icon in the bottom toolbar lets the user drop points and get a real distance reading, turning an abstract 3D scene into concrete, task-relevant numbers. It's tucked into a compact icon-only toolbar, which keeps the main view uncluttered while staying reachable with one thumb.

Part 2 – CG Deconstruction & Reflection



1. LOD % Indicator

The percentage counter (here 72%) reports how much of the current view's mesh tiles have finished streaming at full resolution – a direct window into the CG pipeline as low-res placeholder geometry is swapped for higher-poly detail while the camera is close.

2. Mesh Geometry

From directly above you can see the tower is a reconstructed 3D mesh (via photogrammetry), not a flat texture – the lattice structure has real depth and occludes itself correctly as the camera rotates. This is what lets the renderer compute correct silhouettes and shadows from any angle.

3. Directional Shadow

Even in this near-vertical view, one side of the beams is visibly darker, consistent with a single simulated light source. Shadows are one of the strongest monocular depth cues on a flat screen – they tell the brain which mesh surfaces are raised or recessed.

Reflection

Q1 – Does CG fidelity help or hinder interaction? Realistic lighting/shadows genuinely help spatial orientation at a glance. But there's a tradeoff: when panning or zooming fast, higher-res tiles haven't loaded yet, so you briefly get a blurry low-LOD mesh, which hurts responsiveness and can feel disorienting mid-interaction – fidelity helps static understanding but can hurt fast navigation if tile loading isn't smooth.

Q2 – One concrete HCI improvement. Add a small, non-intrusive loading cue (a subtle shimmer or progress ring) on tiles still streaming in higher resolution, instead of silently popping in sharper geometry. Right now the LOD transition gives no warning, so a small visual cue would reduce the momentary confusion when terrain suddenly sharpens.